

18

$\mathcal{H}_\infty$ Loop Shaping

This chapter introduces a design technique which incorporates loop shaping methods to obtain performance/robust stability tradeoffs, and a particular $\mathcal{H}_\infty$ optimization problem to guarantee closed-loop stability and a level of robust stability at all frequencies. The proposed technique uses only the basic concept of loop shaping methods and then a robust stabilization controller for the normalized coprime factor perturbed system is used to construct the final controller. This chapter is arranged as follows: The $\mathcal{H}_\infty$ theory is applied to solve the stabilization problem of a normalized coprime factor perturbed system in Section 18.1. The loop shaping design procedure is described in Section 18.2. The theoretical justification for the loop shaping design procedure is given in Section 18.3.

18.1 Robust Stabilization of Coprime Factors

In this section, we use the $\mathcal{H}_\infty$ control theory developed in the previous chapters to solve the robust stabilization of a left coprime factor perturbed plant given by

$$P_\Delta = (\tilde{M} + \tilde{\Delta}_M)^{-1}(\tilde{N} + \tilde{\Delta}_N)$$

with $\tilde{M}, \tilde{N}, \tilde{\Delta}_M, \tilde{\Delta}_N \in \mathcal{RH}_\infty$ and $\left\| \begin{bmatrix} \tilde{\Delta}_N & \tilde{\Delta}_M \end{bmatrix} \right\|_\infty < \epsilon$. The transfer matrices $(\tilde{M}, \tilde{N})$ are assumed to be a left coprime factorization of P (i.e., $P = \tilde{M}^{-1}\tilde{N}$), and K internally stabilizes the nominal system.

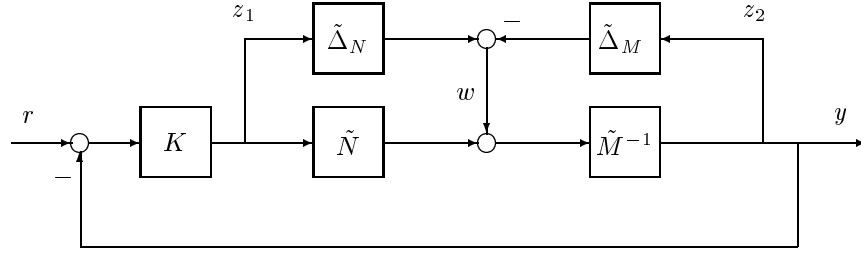


Figure 18.1: Left Coprime Factor Perturbed Systems

It has been shown in Chapter 9 that the system is robustly stable iff

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty \leq 1/\epsilon.$$

Finding a controller such that the above norm condition holds is an $\mathcal{H}_\infty$ norm minimization problem which can be solved using $\mathcal{H}_\infty$ theory developed in the previous chapters.

Suppose P has a stabilizable and detectable state space realization given by

$$P = \left[\begin{array}{c|c} A & B \\ \hline C & D \end{array} \right]$$

and let L be a matrix such that $A + LC$ is stable then a left coprime factorization of $P = \tilde{M}^{-1}\tilde{N}$ is given by

$$\begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} = \left[\begin{array}{c|cc} A + LC & B + LD & L \\ \hline C & D & I \end{array} \right].$$

Denote

$$\hat{K} = -K$$

then the system diagram can be put in an LFT form as in Figure 18.2 with the generalized plant

$$\begin{aligned} G(s) &= \left[\begin{array}{c|c} \begin{bmatrix} 0 \\ \tilde{M}^{-1} \\ \tilde{M}^{-1} \end{bmatrix} & \begin{bmatrix} I \\ P \\ P \end{bmatrix} \end{array} \right] = \left[\begin{array}{c|cc} A & -L & B \\ \hline \begin{bmatrix} 0 \\ C \\ C \end{bmatrix} & \begin{bmatrix} 0 \\ I \\ I \end{bmatrix} & \begin{bmatrix} I \\ D \\ D \end{bmatrix} \end{array} \right] \\ &=: \left[\begin{array}{c|cc} A & B_1 & B_2 \\ \hline C_1 & D_{11} & D_{12} \\ C_2 & D_{21} & D_{22} \end{array} \right]. \end{aligned}$$

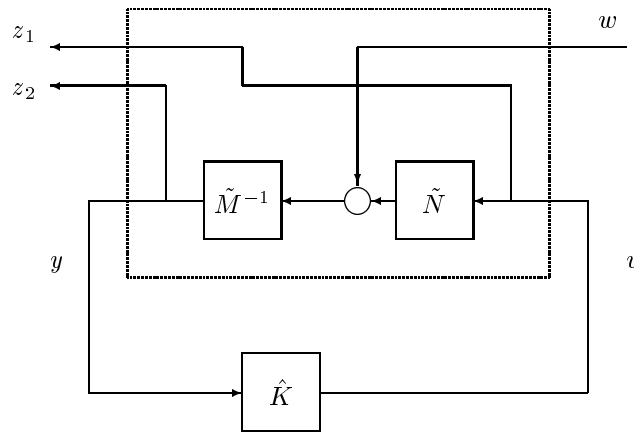


Figure 18.2: LFT Diagram for Coprime Factor Stabilization

To apply the $\mathcal{H}_\infty$ control formulae in Chapter 17, we need to normalize the “ D_{12} ” matrix first. Note that

$$\begin{bmatrix} I \\ D \end{bmatrix} = U \begin{bmatrix} 0 \\ I \end{bmatrix} (I + D^* D)^{\frac{1}{2}}, \quad \text{where } U = \begin{bmatrix} D^*(I + DD^*)^{-\frac{1}{2}} & (I + D^* D)^{-\frac{1}{2}} \\ -(I + DD^*)^{-\frac{1}{2}} & D(I + D^* D)^{-\frac{1}{2}} \end{bmatrix}$$

and U is a unitary matrix. Let

$$\begin{aligned} \hat{K} &= (I + D^*D)^{-\frac{1}{2}} \tilde{K} \\ \begin{bmatrix} z_1 \\ z_2 \end{bmatrix} &= U \begin{bmatrix} \hat{z}_1 \\ \hat{z}_2 \end{bmatrix}. \end{aligned}$$

Then $\|T_{zw}\|_\infty = \|U^*T_{zw}\|_\infty = \|T_{\hat{z}w}\|_\infty$ and the problem becomes of finding a controller $\tilde{K}$ so that $\|T_{\hat{z}w}\|_\infty < \gamma$ with the following generalized plant

$$\hat{G} = \begin{bmatrix} U^* & 0 \\ 0 & I \end{bmatrix} G \begin{bmatrix} I & 0 \\ 0 & (I + D^*D)^{-\frac{1}{2}} \end{bmatrix}$$

$$= \left[\begin{array}{c|cc} A & -L & B \\ \hline \begin{bmatrix} -(I + DD^*)^{-\frac{1}{2}}C \\ (I + D^*D)^{-\frac{1}{2}}D^*C \end{bmatrix} & \begin{bmatrix} -(I + DD^*)^{-\frac{1}{2}} \\ (I + D^*D)^{-\frac{1}{2}}D^* \end{bmatrix} & \begin{bmatrix} 0 \\ I \end{bmatrix} \\ \hline C & I & D(I + D^*D)^{-\frac{1}{2}} \end{array} \right].$$

Now the formulae in Chapter 17 can be applied to $\hat{G}$ to obtain a controller $\tilde{K}$ and then the K can be obtained from $K = -(I + D^*D)^{-\frac{1}{2}}\tilde{K}$. We shall leave the detail to the

reader. In the sequel, we shall consider the case $D = 0$. In this case, we have $\gamma > 1$ and

$$X_\infty \left(A - \frac{LC}{\gamma^2 - 1} \right) + \left(A - \frac{LC}{\gamma^2 - 1} \right)^* X_\infty - X_\infty \left(BB^* - \frac{LL^*}{\gamma^2 - 1} \right) X_\infty + \frac{\gamma^2 C^* C}{\gamma^2 - 1} = 0$$

$$Y_\infty (A + LC)^* + (A + LC) Y_\infty - Y_\infty C^* C Y_\infty = 0.$$

It is clear that $Y_\infty = 0$ is the stabilizing solution. Hence by the formulae in Chapter 17 we have

$$\begin{bmatrix} L_{1\infty} & L_{2\infty} \end{bmatrix} = \begin{bmatrix} 0 & L \end{bmatrix}$$

and

$$Z_\infty = I, \quad \hat{D}_{11} = 0, \quad \hat{D}_{12} = I, \quad \hat{D}_{21} = \frac{\sqrt{\gamma^2 - 1}}{\gamma} I.$$

The results are summarized in the following theorem.

Theorem 18.1 *Let $D = 0$ and let L be such that $A + LC$ is stable then there exists a controller K such that*

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty < \gamma$$

iff $\gamma > 1$ and there exists a stabilizing solution $X_\infty \geq 0$ solving

$$X_\infty \left(A - \frac{LC}{\gamma^2 - 1} \right) + \left(A - \frac{LC}{\gamma^2 - 1} \right)^* X_\infty - X_\infty \left(BB^* - \frac{LL^*}{\gamma^2 - 1} \right) X_\infty + \frac{\gamma^2 C^* C}{\gamma^2 - 1} = 0.$$

Moreover, if the above conditions hold a central controller is given by

$$K = \left[\begin{array}{c|c} A - BB^* X_\infty + LC & L \\ \hline -B^* X_\infty & 0 \end{array} \right].$$

It is clear that the existence of a robust stabilizing controller depends upon the choice of the stabilizing matrix L , i.e., the choice of the coprime factorization. Now let $Y \geq 0$ be the stabilizing solution to

$$AY + YA^* - YC^*CY + BB^* = 0$$

and let $L = -YC^*$. Then the left coprime factorization $(\tilde{M}, \tilde{N})$ given by

$$\begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} = \left[\begin{array}{c|c} A - YC^*C & B \quad -YC^* \\ \hline C & 0 \quad I \end{array} \right]$$

is a normalized left coprime factorization (see Chapter 13).

Corollary 18.2 *Let $D = 0$ and $L = -YC^*$ where $Y \geq 0$ is the stabilizing solution to*

$$AY + YA^* - YC^*CY + BB^* = 0.$$

Then $P = \tilde{M}^{-1}\tilde{N}$ is a normalized left coprime factorization and

$$\begin{aligned} \inf_{K \text{ stabilizing}} \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_{\infty} &= \frac{1}{\sqrt{1 - \lambda_{\max}(YQ)}} \\ &= \left(1 - \left\| \begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} \right\|_H^2 \right)^{-1/2} =: \gamma_{\min} \end{aligned}$$

where Q is the solution to the following Lyapunov equation

$$Q(A - YC^*C) + (A - YC^*C)^*Q + C^*C = 0.$$

Moreover, if the above conditions hold then for any $\gamma > \gamma_{\min}$ a controller achieving

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_{\infty} < \gamma$$

is given by

$$K(s) = \left[\begin{array}{c|c} A - BB^*X_{\infty} - YC^*C & -YC^* \\ \hline -B^*X_{\infty} & 0 \end{array} \right]$$

where

$$X_{\infty} = \frac{\gamma^2}{\gamma^2 - 1} Q \left(I - \frac{\gamma^2}{\gamma^2 - 1} YQ \right)^{-1}.$$

Proof. Note that the Hamiltonian matrix associated with X_{∞} is given by

$$H_{\infty} = \begin{bmatrix} A + \frac{1}{\gamma^2 - 1} YC^*C & -BB^* + \frac{1}{\gamma^2 - 1} YC^*CY \\ -\frac{\gamma^2}{\gamma^2 - 1} C^*C & -(A + \frac{1}{\gamma^2 - 1} YC^*C)^* \end{bmatrix}.$$

Straightforward calculation shows that

$$H_{\infty} = \begin{bmatrix} I & -\frac{\gamma^2}{\gamma^2 - 1} Y \\ 0 & \frac{\gamma^2}{\gamma^2 - 1} I \end{bmatrix} H_q \begin{bmatrix} I & -\frac{\gamma^2}{\gamma^2 - 1} Y \\ 0 & \frac{\gamma^2}{\gamma^2 - 1} I \end{bmatrix}^{-1}$$

where

$$H_q = \begin{bmatrix} A - YC^*C & 0 \\ -C^*C & -(A - YC^*C)^* \end{bmatrix}.$$

It is clear that the stable invariant subspace of H_q is given by

$$\mathcal{X}_-(H_q) = \text{Im} \begin{bmatrix} I \\ Q \end{bmatrix}$$

and the stable invariant subspace of H_∞ is given by

$$\mathcal{X}_-(H_\infty) = \begin{bmatrix} I & -\frac{\gamma^2}{\gamma^2-1}Y \\ 0 & \frac{\gamma^2}{\gamma^2-1}I \end{bmatrix} \mathcal{X}_-(H_q) = \text{Im} \begin{bmatrix} I - \frac{\gamma^2}{\gamma^2-1}YQ \\ \frac{\gamma^2}{\gamma^2-1}Q \end{bmatrix}.$$

Hence there is a nonnegative definite stabilizing solution to the algebraic Riccati equation of X_∞ if and only if

$$I - \frac{\gamma^2}{\gamma^2-1}YQ > 0$$

or

$$\gamma > \frac{1}{\sqrt{1 - \lambda_{\max}(YQ)}}$$

and the solution, if it exists, is given by

$$X_\infty = \frac{\gamma^2}{\gamma^2-1}Q \left(I - \frac{\gamma^2}{\gamma^2-1}YQ \right)^{-1}.$$

Note that Y and Q are the controllability Gramian and the observability Gramian of $\begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix}$ respectively. Therefore, we also have that the Hankel norm of $\begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix}$ is $\sqrt{\lambda_{\max}(YQ)}$. $\square$

Corollary 18.3 *Let $P = \tilde{M}^{-1}\tilde{N}$ be a normalized left coprime factorization and*

$$P_\Delta = (\tilde{M} + \tilde{\Delta}_M)^{-1}(\tilde{N} + \tilde{\Delta}_N)$$

with

$$\left\| \begin{bmatrix} \tilde{\Delta}_N & \tilde{\Delta}_M \end{bmatrix} \right\|_\infty < \epsilon.$$

Then there is a robustly stabilizing controller for P_Δ if and only if

$$\epsilon \leq \sqrt{1 - \lambda_{\max}(YQ)} = \sqrt{1 - \left\| \begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} \right\|_H^2}.$$

The solutions to the normalized left coprime factorization stabilization problem are also solutions to a related $\mathcal{H}_\infty$ problem which is shown in the following lemma.

Lemma 18.4 *Let $P = \tilde{M}^{-1}\tilde{N}$ be a normalized left coprime factorization. Then*

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty = \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix} \right\|_\infty.$$

Proof. Since $(\tilde{M}, \tilde{N})$ is a normalized left coprime factorization of P , we have

$$\begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}^\sim = I$$

and

$$\left\| \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} \right\|_\infty = \left\| \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}^\sim \right\|_\infty = 1.$$

Using these equations, we have

$$\begin{aligned} & \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty \\ &= \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}^\sim \right\|_\infty \\ &\leq \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} \right\|_\infty \left\| \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix}^\sim \right\|_\infty \\ &= \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix} \right\|_\infty \\ &\leq \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty \left\| \begin{bmatrix} \tilde{M} & \tilde{N} \end{bmatrix} \right\|_\infty \\ &= \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty. \end{aligned}$$

This implies

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_\infty = \left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix} \right\|_\infty.$$

□

Corollary 18.5 *A controller solves the normalized left coprime factor robust stabilization problem if and only if it solves the following $\mathcal{H}_\infty$ control problem*

$$\left\| \begin{bmatrix} I \\ K \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix} \right\|_\infty < \gamma$$

and

$$\begin{aligned} \inf_{K \text{ stabilizing}} \left\| \begin{bmatrix} I \\ K \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix} \right\|_\infty &= \frac{1}{\sqrt{1 - \lambda_{\max}(YQ)}} \\ &= \left(1 - \left\| \begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix} \right\|_H^2 \right)^{-1/2}. \end{aligned}$$

The solution Q can also be obtained in other ways. Let $X \geq 0$ be the stabilizing solution to

$$XA + A^*X - XBB^*X + C^*C = 0$$

then it is easy to verify that

$$Q = (I + XY)^{-1}X.$$

Hence

$$\gamma_{\min} = \frac{1}{\sqrt{1 - \lambda_{\max}(YQ)}} = \left(1 - \left\| \begin{bmatrix} \tilde{N} & \tilde{M} \end{bmatrix}_H \right\|^2\right)^{-1/2} = \sqrt{1 + \lambda_{\max}(XY)}.$$

Similar results can be obtained if one starts with a normalized right coprime factorization. In fact, a rather strong relation between the normalized left and right coprime factorization problems can be established using the following matrix fact.

Lemma 18.6 *Let M and N be any compatibly dimensioned complex matrices such that $MM = M$, $NN = N$, and $M + N = I$. Then $\sigma_i(M) = \sigma_i(N)$ for all i such that $0 < \sigma_i(M) \neq 1$.*

Proof. We first show that the eigenvalues of M and N are either 0 or 1 and M and N are diagonalizable. In fact, assume that λ is an eigenvalue of M and x is a corresponding eigenvector, then $\lambda x = Mx = MMx = M(Mx) = \lambda Mx = \lambda^2 x$, i.e., $\lambda(1 - \lambda)x = 0$. This implies that either $\lambda = 0$ or $\lambda = 1$. To show that M is diagonalizable, assume $M = TJT^{-1}$ where J is a Jordan canonical form, it follows immediately that J must be diagonal by the condition $M = MM$. The proof for N is similar.

Next, assume that M is diagonalized by a nonsingular matrix T such that

$$M = T \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} T^{-1}.$$

Then

$$N = I - M = T \begin{bmatrix} 0 & 0 \\ 0 & I \end{bmatrix} T^{-1}.$$

Define

$$\begin{bmatrix} A & B \\ B^* & D \end{bmatrix} := T^*T$$

and assume $0 < \lambda \neq 1$. Then $A > 0$ and

$$\begin{aligned} \det(M^*M - \lambda I) &= 0 \\ \Leftrightarrow \det\left(\begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} T^*T \begin{bmatrix} I & 0 \\ 0 & 0 \end{bmatrix} - T^*T\right) &= 0 \end{aligned}$$

$$\begin{aligned}
&\Leftrightarrow \det \begin{bmatrix} (1-\lambda)A & -\lambda B \\ -\lambda B^* & -\lambda D \end{bmatrix} = 0 \\
&\Leftrightarrow \det(-\lambda D - \frac{\lambda^2}{1-\lambda} B^* A^{-1} B) = 0 \\
&\Leftrightarrow \det(\frac{1-\lambda}{\lambda} D + B^* A^{-1} B) = 0 \\
&\Leftrightarrow \det \begin{bmatrix} -\lambda A & -\lambda B \\ -\lambda B^* & (1-\lambda)D \end{bmatrix} = 0 \\
&\Leftrightarrow \det(N^* N - \lambda I) = 0.
\end{aligned}$$

This implies that all nonzero eigenvalues of $M^* M$ and $N^* N$ that are not equal to 1 are equal, i.e., $\sigma_i(M) = \sigma_i(N)$ for all i such that $0 < \sigma_i(M) \neq 1$. $\square$

Using this matrix fact, we have the following corollary.

Corollary 18.7 *Let K and P be any compatibly dimensioned complex matrices. Then*

$$\left\| \begin{bmatrix} I \\ K \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix} \right\| = \left\| \begin{bmatrix} I \\ P \end{bmatrix} (I + KP)^{-1} \begin{bmatrix} I & K \end{bmatrix} \right\|.$$

Proof. Define

$$M = \begin{bmatrix} I \\ K \end{bmatrix} (I + PK)^{-1} \begin{bmatrix} I & P \end{bmatrix}, \quad N = \begin{bmatrix} -P \\ I \end{bmatrix} (I + KP)^{-1} \begin{bmatrix} -K & I \end{bmatrix}.$$

Then it is easy to verify that $MM = M$, $NN = N$ and $M + N = I$. By Lemma 18.6, we have $\|M\| = \|N\|$. The corollary follows by noting that

$$\begin{bmatrix} I \\ P \end{bmatrix} (I + KP)^{-1} \begin{bmatrix} I & K \end{bmatrix} = \begin{bmatrix} 0 & I \\ -I & 0 \end{bmatrix} N \begin{bmatrix} 0 & -I \\ I & 0 \end{bmatrix}.$$

$\square$

Corollary 18.8 *Let $P = \tilde{M}^{-1} \tilde{N} = N M^{-1}$ be respectively the normalized left and right coprime factorizations. Then*

$$\left\| \begin{bmatrix} K \\ I \end{bmatrix} (I + PK)^{-1} \tilde{M}^{-1} \right\|_{\infty} = \left\| M^{-1} (I + KP)^{-1} \begin{bmatrix} I & K \end{bmatrix} \right\|_{\infty}.$$

Proof. This follows from Corollary 18.7 and the fact that

$$\left\| M^{-1}(I + KP)^{-1} \begin{bmatrix} I & K \end{bmatrix} \right\|_\infty = \left\| \begin{bmatrix} I \\ P \end{bmatrix} (I + KP)^{-1} \begin{bmatrix} I & K \end{bmatrix} \right\|_\infty.$$

□

This corollary says that any $\mathcal{H}_\infty$ controller for the normalized left coprime factorization is also an $\mathcal{H}_\infty$ controller for the normalized right coprime factorization. Hence one can work with either factorization.

18.2 Loop Shaping Using Normalized Coprime Stabilization

This section considers the $\mathcal{H}_\infty$ loop shaping design. The objective of this approach is to incorporate the simple performance/robustness tradeoff obtained in the loop shaping, with the guaranteed stability properties of $\mathcal{H}_\infty$ design methods. Recall from Section 5.5 of Chapter 5 that good performance controller design requires that

$$\bar{\sigma}((I + PK)^{-1}), \quad \bar{\sigma}((I + PK)^{-1}P), \quad \bar{\sigma}((I + KP)^{-1}), \quad \bar{\sigma}(K(I + PK)^{-1}) \quad (18.1)$$

be made small, particularly in some low frequency range. And good robustness requires that

$$\bar{\sigma}(PK(I + PK)^{-1}), \quad \bar{\sigma}(KP(I + KP)^{-1}) \quad (18.2)$$

be made small, particularly in some high frequency range. These requirements in turn imply that good controller design boils down to achieving the desired loop (and controller) gains in the appropriate frequency range:

$$\underline{\sigma}(PK) \gg 1, \quad \underline{\sigma}(KP) \gg 1, \quad \underline{\sigma}(K) \gg 1$$

in some low frequency range and

$$\bar{\sigma}(PK) \ll 1, \quad \bar{\sigma}(KP) \ll 1, \quad \bar{\sigma}(K) \leq M$$

in some high frequency range where M is not too large.

The design procedure is stated below.

Loop Shaping Design Procedure

- (1) Loop Shaping: Using a precompensator W_1 and/or a postcompensator W_2 , the singular values of the nominal plant are shaped to give a desired open-loop shape. The nominal plant G and the shaping functions W_1, W_2 are combined to form the shaped plant, G_s where $G_s = W_2 G W_1$. We assume that W_1 and W_2 are such that G_s contains no hidden modes.

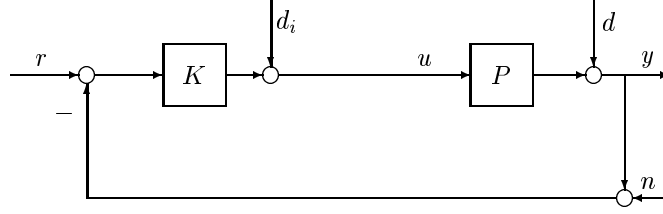


Figure 18.3: Standard Feedback Configuration

- (2) Robust Stabilization: a) Calculate ϵ_{max} , where

$$\begin{aligned}\epsilon_{max} &= \left(\inf_{K \text{ stabilizing}} \left\| \begin{bmatrix} I \\ K \end{bmatrix} (I + G_s K)^{-1} \tilde{M}_s^{-1} \right\|_{\infty} \right)^{-1} \\ &= \sqrt{1 - \left\| \begin{bmatrix} \tilde{N}_s & \tilde{M}_s \end{bmatrix} \right\|_H^2} < 1\end{aligned}$$

and $\tilde{M}_s, \tilde{N}_s$ define the normalized coprime factors of G_s such that $G_s = \tilde{M}_s^{-1} \tilde{N}_s$ and

$$\tilde{M}_s \tilde{M}_s^{\sim} + \tilde{N}_s \tilde{N}_s^{\sim} = I.$$

If $\epsilon_{max} \ll 1$ return to (1) and adjust W_1 and W_2 .

- b) Select $\epsilon \leq \epsilon_{max}$, then synthesize a stabilizing controller K_{∞} , which satisfies

$$\left\| \begin{bmatrix} I \\ K_{\infty} \end{bmatrix} (I + G_s K_{\infty})^{-1} \tilde{M}_s^{-1} \right\|_{\infty} \leq \epsilon^{-1}.$$

- (3) The final feedback controller K is then constructed by combining the $\mathcal{H}_{\infty}$ controller K_{∞} with the shaping functions W_1 and W_2 such that

$$K = W_1 K_{\infty} W_2.$$

A typical design works as follows: the designer inspects the open-loop singular values of the nominal plant, and shapes these by pre- and/or postcompensation until nominal performance (and possibly robust stability) specifications are met. (Recall that the open-loop shape is related to closed-loop objectives.) A feedback controller K_{∞} with associated stability margin (for the shaped plant) $\epsilon \leq \epsilon_{max}$, is then synthesized. If ϵ_{max} is small, then the specified loop shape is incompatible with robust stability requirements, and should be adjusted accordingly, then K_{∞} is reevaluated.

In the above design procedure we have specified the desired loop shape by $W_2 G W_1$. But, after Stage (2) of the design procedure, the actual loop shape achieved is in fact

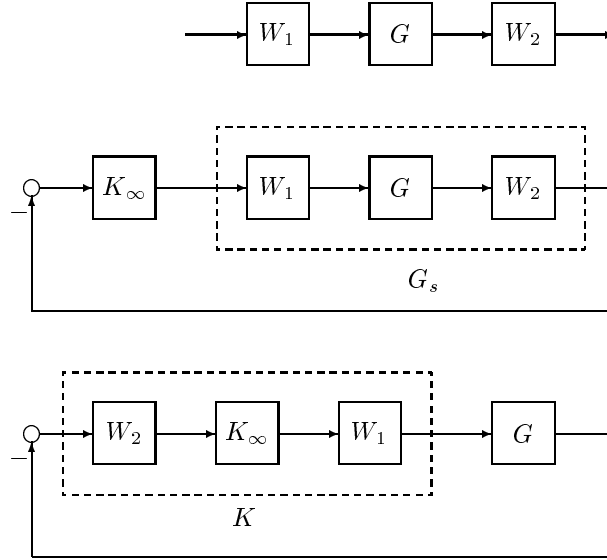


Figure 18.4: The Loop Shaping Design Procedure

given by $W_1 K_\infty W_2 G$ at plant input and $G W_1 K_\infty W_2$ at plant output. It is therefore possible that the inclusion of K_∞ in the open-loop transfer function will cause deterioration in the open-loop shape specified by G_s . In the next section, we will show that the degradation in the loop shape caused by the $\mathcal{H}_\infty$ controller K_∞ is limited at frequencies where the specified loop shape is sufficiently large or sufficiently small. In particular, we show in the next section that ϵ can be interpreted as an indicator of the success of the loop shaping in addition to providing a robust stability guarantee for the closed-loop systems. A small value of ϵ_{max} ($\epsilon_{max} \ll 1$) in Stage (2) always indicates incompatibility between the specified loop shape, the nominal plant phase, and robust closed-loop stability.

Remark 18.1 Note that, in contrast to the classical loop shaping approach, the loop shaping here is done without explicit regard for the nominal plant phase information. That is, closed-loop stability requirements are disregarded at this stage. Also, in contrast with conventional $\mathcal{H}_\infty$ design, the robust stabilization is done without frequency weighting. The design procedure described here is both simple and systematic, and only assumes knowledge of elementary loop shaping principles on the part of the designer.

♡

Remark 18.2 The above robust stabilization objective can also be interpreted as the more standard $\mathcal{H}_\infty$ problem formulation of minimizing the $\mathcal{H}_\infty$ norm of the frequency weighted gain from disturbances on the plant input and output to the controller input

and output as follows.

$$\begin{aligned}
\left\| \begin{bmatrix} I \\ K_\infty \end{bmatrix} (I + G_s K_\infty)^{-1} \tilde{M}_s^{-1} \right\|_\infty &= \left\| \begin{bmatrix} I \\ K_\infty \end{bmatrix} (I + G_s K_\infty)^{-1} \begin{bmatrix} I & G_s \end{bmatrix} \right\|_\infty \\
&= \left\| \begin{bmatrix} W_2 \\ W_1^{-1} K \end{bmatrix} (I + GK)^{-1} \begin{bmatrix} W_2^{-1} & GW_1 \end{bmatrix} \right\|_\infty \\
&= \left\| \begin{bmatrix} I \\ G_s \end{bmatrix} (I + K_\infty G_s)^{-1} \begin{bmatrix} I & K_\infty \end{bmatrix} \right\|_\infty \\
&= \left\| \begin{bmatrix} W_1^{-1} \\ W_2 G \end{bmatrix} (I + KG)^{-1} \begin{bmatrix} W_1 & GW_2^{-1} \end{bmatrix} \right\|_\infty
\end{aligned}$$

This shows how all the closed-loop objectives in (18.1) and (18.2) are incorporated. $\heartsuit$

18.3 Theoretical Justification for $\mathcal{H}_\infty$ Loop Shaping

The objective of this section is to provide justification for the use of parameter ϵ as a design indicator. We will show that ϵ is a measure of both closed-loop robust stability and the success of the design in meeting the loop shaping specifications.

We first examine the possibility of loop shape deterioration at frequencies of high loop gain (typically low frequency). At low frequency (in particular, $\omega \in (0, \omega_l)$), the deterioration in loop shape at plant output can be obtained by comparing $\underline{\sigma}(GW_1 K_\infty W_2)$ to $\underline{\sigma}(G_s) = \underline{\sigma}(W_2 G W_1)$. Note that

$$\underline{\sigma}(GK) = \underline{\sigma}(GW_1 K_\infty W_2) = \underline{\sigma}(W_2^{-1} W_2 G W_1 K_\infty W_2) \geq \underline{\sigma}(W_2 G W_1) \underline{\sigma}(K_\infty) / \kappa(W_2) \quad (18.3)$$

where $\kappa(\cdot)$ denotes condition number. Similarly, for loop shape deterioration at plant input, we have

$$\underline{\sigma}(KG) = \underline{\sigma}(W_1 K_\infty W_2 G) = \underline{\sigma}(W_1 K_\infty W_2 G W_1 W_1^{-1}) \geq \underline{\sigma}(W_2 G W_1) \underline{\sigma}(K_\infty) / \kappa(W_1). \quad (18.4)$$

In each case, $\underline{\sigma}(K_\infty)$ is required to obtain a bound on the deterioration in the loop shape at low frequency. Note that the condition numbers $\kappa(W_1)$ and $\kappa(W_2)$ are selected by the designer.

Next, recalling that G_s denotes the shaped plant, and that K_∞ robustly stabilizes the normalized coprime factorization of G_s with stability margin ϵ , then we have

$$\left\| \begin{bmatrix} I \\ K_\infty \end{bmatrix} (I + G_s K_\infty)^{-1} \tilde{M}_s^{-1} \right\|_\infty \leq \epsilon^{-1} := \gamma \quad (18.5)$$

where $(\tilde{N}_s, \tilde{M}_s)$ is a normalized left coprime factorization of G_s , and the parameter γ is defined to simplify the notation to follow. The following result shows that $\underline{\sigma}(K_\infty)$

is explicitly bounded by functions of ϵ and $\underline{\sigma}(G_s)$, the minimum singular value of the shaped plant, and hence by (18.3) and (18.4) K_∞ will only have a limited effect on the specified loop shape at low frequency.

Theorem 18.9 *Any controller K_∞ satisfying (18.5), where G_s is assumed square, also satisfies*

$$\underline{\sigma}(K_\infty(j\omega)) \geq \frac{\underline{\sigma}(G_s(j\omega)) - \sqrt{\gamma^2 - 1}}{\sqrt{\gamma^2 - 1}\underline{\sigma}(G_s(j\omega)) + 1}$$

for all ω such that

$$\underline{\sigma}(G_s(j\omega)) > \sqrt{\gamma^2 - 1}.$$

Furthermore, if $\underline{\sigma}(G_s) \gg \sqrt{\gamma^2 - 1}$, then $\underline{\sigma}(K_\infty(j\omega)) \gtrsim 1/\sqrt{\gamma^2 - 1}$, where $\gtrsim$ denotes asymptotically greater than or equal to as $\underline{\sigma}(G_s) \rightarrow \infty$.

Proof. First note that $\underline{\sigma}(G_s) > \sqrt{\gamma^2 - 1}$ implies that

$$I + G_s G_s^* > \gamma^2 I.$$

Further since $(\tilde{N}_s, \tilde{M}_s)$ is a normalized left coprime factorization of G_s , we have

$$\tilde{M}_s \tilde{M}_s^* = I - \tilde{N}_s \tilde{N}_s^* = I - \tilde{M}_s G_s G_s^* \tilde{M}_s^*.$$

Then

$$\tilde{M}_s^* \tilde{M}_s = (I + G_s G_s^*)^{-1} < \gamma^{-2} I.$$

Now

$$\left\| \begin{bmatrix} I \\ K_\infty \end{bmatrix} (I + G_s K_\infty)^{-1} \tilde{M}_s^{-1} \right\|_\infty \leq \gamma$$

can be rewritten as

$$(I + K_\infty^* K_\infty) \leq \gamma^2 (I - K_\infty^* G_s^*) (\tilde{M}_s^* \tilde{M}_s) (I - G_s K_\infty). \quad (18.6)$$

We will next show that K_∞ is invertible. Suppose that there exists an x such that $K_\infty x = 0$, then $x^* \times (18.6) \times x$ gives

$$\gamma^{-2} x^* x \leq x^* \tilde{M}_s^* \tilde{M}_s x$$

which implies that $x = 0$ since $\tilde{M}_s^* \tilde{M}_s < \gamma^{-2} I$, and hence K_∞ is invertible. Equation (18.6) can now be written as

$$(K_\infty^{-*} K_\infty^{-1} + I) \leq \gamma^2 (K_\infty^{-*} - G_s^*) \tilde{M}_s^* \tilde{M}_s (K_\infty^{-1} - G_s). \quad (18.7)$$

Define W such that

$$(W W^*)^{-1} = I - \gamma^2 \tilde{M}_s^* \tilde{M}_s = I - \gamma^2 (I + G_s G_s^*)^{-1}$$

and completing the square in (18.7) with respect to K_∞^{-1} yields

$$(K_\infty^{-*} + N^*)(WW^*)^{-1}(K_\infty^{-1} + N) \leq (\gamma^2 - 1)R^*R$$

where

$$\begin{aligned} N &= \gamma^2 G_s((1 - \gamma^2)I + G_s^* G_s)^{-1} \\ R^* R &= (I + G_s^* G_s)((1 - \gamma^2)I + G_s^* G_s)^{-1}. \end{aligned}$$

Hence we have

$$R^{-*}(K_\infty^{-*} + N^*)(WW^*)^{-1}(K_\infty^{-1} + N)R^{-1} \leq (\gamma^2 - 1)I$$

and

$$\bar{\sigma}(W^{-1}(K_\infty^{-1} + N)R^{-1}) \leq \sqrt{\gamma^2 - 1}.$$

Use $\bar{\sigma}(W^{-1}(K_\infty^{-1} + N)R^{-1}) \geq \underline{\sigma}(W^{-1})\bar{\sigma}(K_\infty^{-1} + N)\underline{\sigma}(R^{-1})$ to get

$$\bar{\sigma}(K_\infty^{-1} + N) \leq \sqrt{\gamma^2 - 1}\bar{\sigma}(W)\bar{\sigma}(R)$$

and use $\underline{\sigma}(K_\infty^{-1} + N) \geq \underline{\sigma}(K_\infty) - \bar{\sigma}(N)$ to get

$$\underline{\sigma}(K_\infty) \geq \left\{ (\gamma^2 - 1)^{1/2}\bar{\sigma}(W)\bar{\sigma}(R) + \bar{\sigma}(N) \right\}^{-1}. \quad (18.8)$$

Next, note that the eigenvalues of WW^* , N^*N , and R^*R can be computed as follows

$$\begin{aligned} \lambda(WW^*) &= \frac{1 + \lambda(G_s G_s^*)}{1 - \gamma^2 + \lambda(G_s G_s^*)} \\ \lambda(N^*N) &= \frac{\gamma^4 \lambda(G_s G_s^*)}{(1 - \gamma^2 + \lambda(G_s G_s^*))^2} \\ \lambda(R^*R) &= \frac{1 + \lambda(G_s G_s^*)}{1 - \gamma^2 + \lambda(G_s G_s^*)} \end{aligned}$$

therefore

$$\begin{aligned} \bar{\sigma}(W) &= \sqrt{\lambda_{\max}(WW^*)} = \left(\frac{1 + \lambda_{\min}(G_s G_s^*)}{1 - \gamma^2 + \lambda_{\min}(G_s G_s^*)} \right)^{1/2} = \left(\frac{1 + \underline{\sigma}^2(G_s)}{1 - \gamma^2 + \underline{\sigma}^2(G_s)} \right)^{1/2} \\ \bar{\sigma}(N) &= \sqrt{\lambda_{\max}(N^*N)} = \frac{\gamma^2 \sqrt{\lambda_{\min}(G_s G_s^*)}}{1 - \gamma^2 + \lambda_{\min}(G_s G_s^*)} = \frac{\gamma^2 \underline{\sigma}(G_s)}{1 - \gamma^2 + \underline{\sigma}^2(G_s)} \\ \bar{\sigma}(R) &= \sqrt{\lambda_{\max}(R^*R)} = \left(\frac{1 + \lambda_{\min}(G_s G_s^*)}{1 - \gamma^2 + \lambda_{\min}(G_s G_s^*)} \right)^{1/2} = \left(\frac{1 + \underline{\sigma}^2(G_s)}{1 - \gamma^2 + \underline{\sigma}^2(G_s)} \right)^{1/2}. \end{aligned}$$

Substituting these formulas into (18.8), we have

$$\underline{\sigma}(K_\infty) \geq \left\{ \frac{(\gamma^2 - 1)^{1/2}(1 + \underline{\sigma}^2(G_s)) + \gamma^2 \underline{\sigma}(G_s)}{\underline{\sigma}^2(G_s) - (\gamma^2 - 1)} \right\}^{-1} = \frac{\underline{\sigma}(G_s) - \sqrt{\gamma^2 - 1}}{\sqrt{\gamma^2 - 1} \underline{\sigma}(G_s) + 1}.$$

□

The main implication of Theorem 18.9 is that the bound on $\underline{\sigma}(K_\infty)$ depends only on the selected loop shape, and the stability margin of the shaped plant. The value of $\gamma (= \epsilon^{-1})$ directly determines the frequency range over which this result is valid—a small γ (large ϵ) is desirable, as we would expect. Further, G_s has a sufficiently large loop gain, then so also will $G_s K_\infty$ provided $\gamma (= \epsilon^{-1})$ is sufficiently small.

In an analogous manner, we now examine the possibility of deterioration in the loop shape at high frequency due to the inclusion of K_∞ . Note that at high frequency (in particular, $\omega \in (\omega_h, \infty)$) the deterioration in plant output loop shape can be obtained by comparing $\bar{\sigma}(GW_1 K_\infty W_2)$ to $\bar{\sigma}(G_s) = \bar{\sigma}(W_2 G W_1)$. Note that, analogously to (18.3) and (18.4) we have

$$\bar{\sigma}(GK) = \bar{\sigma}(GW_1 K_\infty W_2) \leq \bar{\sigma}(W_2 G W_1) \bar{\sigma}(K_\infty) \kappa(W_2).$$

Similarly, the corresponding deterioration in plant input loop shape is obtained by comparing $\bar{\sigma}(W_1 K_\infty W_2 G)$ to $\bar{\sigma}(W_2 G W_1)$ where

$$\bar{\sigma}(KG) = \bar{\sigma}(W_1 K_\infty W_2 G) \leq \bar{\sigma}(W_2 G W_1) \bar{\sigma}(K_\infty) \kappa(W_1).$$

Hence, in each case, $\bar{\sigma}(K_\infty)$ is required to obtain a bound on the deterioration in the loop shape at high frequency. In an identical manner to Theorem 18.9, we now show that $\bar{\sigma}(K_\infty)$ is explicitly bounded by functions of γ , and $\bar{\sigma}(G_s)$, the maximum singular value of the shaped plant.

Theorem 18.10 *Any controller K_∞ satisfying (18.5) also satisfies*

$$\underline{\sigma}(K_\infty(j\omega)) \leq \frac{\sqrt{\gamma^2 - 1} + \bar{\sigma}(G_s(j\omega))}{1 - \sqrt{\gamma^2 - 1} \bar{\sigma}(G_s(j\omega))}$$

for all ω such that

$$\bar{\sigma}(G_s(j\omega)) < \frac{1}{\sqrt{\gamma^2 - 1}}.$$

Furthermore, if $\bar{\sigma}(G_s) \ll 1/\sqrt{\gamma^2 - 1}$, then $\underline{\sigma}(K_\infty(j\omega)) \lesssim \sqrt{\gamma^2 - 1}$, where $\lesssim$ denotes asymptotically less than or equal to as $\bar{\sigma}(G_s) \rightarrow 0$.

Proof. The proof of Theorem 18.10 is similar to that of Theorem 18.9, and is only sketched here: As in the proof of Theorem 18.9, we have $\tilde{M}_s^* \tilde{M}_s = (I + G_s G_s^*)^{-1}$ and

$$(I + K_\infty^* K_\infty) \leq \gamma^2 (I - K_\infty^* G_s^*) (\tilde{M}_s^* \tilde{M}_s) (I - G_s K_\infty). \quad (18.9)$$

Since $\bar{\sigma}(G_s) < \frac{1}{\sqrt{\gamma^2 - 1}}$,

$$I - \gamma^2 G_s^* (I + G_s G_s^*)^{-1} G_s > 0$$

and there exists a spectral factorization

$$V^* V = I - \gamma^2 G_s^* (I + G_s G_s^*)^{-1} G_s.$$

Now completing the square in (18.9) with respect to K_∞ yields

$$(K_\infty^* + M^*) V^* V (K_\infty + M) \leq (\gamma^2 - 1) Y^* Y$$

where

$$\begin{aligned} M &= \gamma^2 G_s^* (I + (1 - \gamma^2) G_s G_s^*)^{-1} \\ Y^* Y &= (\gamma^2 - 1) (I + G_s G_s^*) (I + (1 - \gamma^2) G_s G_s^*)^{-1}. \end{aligned}$$

Hence we have

$$\bar{\sigma}(V(K_\infty + M)Y^{-1}) \leq \sqrt{\gamma^2 - 1}$$

which implies

$$\bar{\sigma}(K_\infty) \leq \frac{\sqrt{\gamma^2 - 1}}{\underline{\sigma}(V)\underline{\sigma}(Y^{-1})} + \bar{\sigma}(M). \quad (18.10)$$

As in the proof of Theorem 18.9, it is easy to show that

$$\begin{aligned} \underline{\sigma}(V) &= \underline{\sigma}(Y^{-1}) = \left(\frac{1 - (\gamma^2 - 1)\bar{\sigma}^2(G_s)}{1 + \bar{\sigma}^2(G_s)} \right)^{1/2} \\ \bar{\sigma}(M) &= \frac{\gamma^2 \bar{\sigma}(G_s)}{1 - (\gamma^2 - 1)\bar{\sigma}^2(G_s)}. \end{aligned}$$

Substituting these formulas into (18.10), we have

$$\bar{\sigma}(K_\infty) \leq \frac{(\gamma^2 - 1)^{1/2}(1 + \bar{\sigma}^2(G_s)) + \gamma^2 \bar{\sigma}(G_s)}{1 - (\gamma^2 - 1)\bar{\sigma}^2(G_s)} = \frac{\sqrt{\gamma^2 - 1} + \bar{\sigma}(G_s)}{1 - \sqrt{\gamma^2 - 1}\bar{\sigma}(G_s)}.$$

□

The results in Theorem 18.9 and 18.10 confirm that γ (alternatively ϵ) indicates the compatibility between the specified loop shape and closed-loop stability requirements.

Theorem 18.11 *Let G be the nominal plant and let $K = W_1 K_\infty W_2$ be the associated controller obtained from loop shaping design procedure in the last section. Then if*

$$\left\| \begin{bmatrix} I \\ K_\infty \end{bmatrix} (I + G_s K_\infty)^{-1} \tilde{M}_s^{-1} \right\|_\infty \leq \gamma$$

we have

$$\bar{\sigma}(K(I + GK)^{-1}) \leq \gamma \bar{\sigma}(\tilde{M}_s) \bar{\sigma}(W_1) \bar{\sigma}(W_2) \quad (18.11)$$

$$\bar{\sigma}((I + GK)^{-1}) \leq \min \left\{ \gamma \bar{\sigma}(\tilde{M}_s) \kappa(W_2), 1 + \gamma \bar{\sigma}(N_s) \kappa(W_2) \right\} \quad (18.12)$$

$$\bar{\sigma}(K(I + GK)^{-1}G) \leq \min \left\{ \gamma \bar{\sigma}(\tilde{N}_s) \kappa(W_1), 1 + \gamma \bar{\sigma}(M_s) \kappa(W_1) \right\} \quad (18.13)$$

$$\bar{\sigma}((I + GK)^{-1}G) \leq \frac{\gamma \bar{\sigma}(\tilde{N}_s)}{\underline{\sigma}(W_1) \underline{\sigma}(W_2)} \quad (18.14)$$

$$\bar{\sigma}((I + KG)^{-1}) \leq \min \left\{ 1 + \gamma \bar{\sigma}(\tilde{N}_s) \kappa(W_1), \gamma \bar{\sigma}(M_s) \kappa(W_1) \right\} \quad (18.15)$$

$$\bar{\sigma}(G(I + KG)^{-1}K) \leq \min \left\{ 1 + \gamma \bar{\sigma}(\tilde{M}_s) \kappa(W_2), \gamma \bar{\sigma}(N_s) \kappa(W_2) \right\} \quad (18.16)$$

where

$$\bar{\sigma}(\tilde{N}_s) = \bar{\sigma}(N_s) = \left(\frac{\bar{\sigma}^2(W_2 G W_1)}{1 + \bar{\sigma}^2(W_2 G W_1)} \right)^{1/2} \quad (18.17)$$

$$\bar{\sigma}(\tilde{M}_s) = \bar{\sigma}(M_s) = \left(\frac{1}{1 + \bar{\sigma}^2(W_2 G W_1)} \right)^{1/2} \quad (18.18)$$

and $(\tilde{N}_s, \tilde{M}_s)$, respectively, (N_s, M_s) , is a normalized left coprime factorization, respectively, right coprime factorization, of $G_s = W_2 G W_1$.

Proof. Note that

$$\tilde{M}_s^* \tilde{M}_s = (I + G_s G_s^*)^{-1}$$

and

$$\tilde{M}_s \tilde{M}_s^* = I - \tilde{N}_s \tilde{N}_s^*.$$

Then

$$\begin{aligned} \bar{\sigma}^2(\tilde{M}_s) &= \lambda_{\max}(\tilde{M}_s^* \tilde{M}_s) = \frac{1}{1 + \lambda_{\max}(G_s G_s^*)} = \frac{1}{1 + \bar{\sigma}^2(G_s)} \\ \bar{\sigma}^2(\tilde{N}_s) &= 1 - \bar{\sigma}^2(M_s) = \frac{\bar{\sigma}^2(G_s)}{1 + \bar{\sigma}^2(G_s)}. \end{aligned}$$

The proof for the normalized right coprime factorization is similar. All other inequalities follow from noting

$$\left\| \begin{bmatrix} I \\ K_\infty \end{bmatrix} (I + G_s K_\infty)^{-1} \tilde{M}_s^{-1} \right\|_\infty \leq \gamma$$

and

$$\begin{aligned} \left\| \begin{bmatrix} I \\ K_\infty \end{bmatrix} (I + G_s K_\infty)^{-1} \tilde{M}_s^{-1} \right\|_\infty &= \left\| \begin{bmatrix} W_2 \\ W_1^{-1} K \end{bmatrix} (I + GK)^{-1} \begin{bmatrix} W_2^{-1} & GW_1 \end{bmatrix} \right\|_\infty \\ &= \left\| \begin{bmatrix} W_1^{-1} \\ W_2 G \end{bmatrix} (I + KG)^{-1} \begin{bmatrix} W_1 & GW_2^{-1} \end{bmatrix} \right\|_\infty \end{aligned}$$

□

This theorem shows that all closed-loop objectives are guaranteed to have bounded magnitude and the bounds depend only on γ , W_1 , W_2 , and G .

18.4 Notes and References

The $\mathcal{H}_\infty$ loop shaping using normalized coprime factorization was developed by McFarlane and Glover [1990, 1992]. In the same references, some design examples were also shown. The method has been applied to the design of scheduled controllers for a VSTOL aircraft in Hyde and Glover [1993]. The robust stabilization of normalized coprime factors is closely related to the robustness in the gap metric and graph topology, see El-Sakkary [1985], Georgiou and Smith [1990], Glover and McFarlane [1989], McFarlane, Glover, and Vidyasagar [1990], Qiu and Davison [1992a, 1992b], Vinnicombe [1993], Vidyasagar [1984, 1985], Zhu [1989], and references therein.

